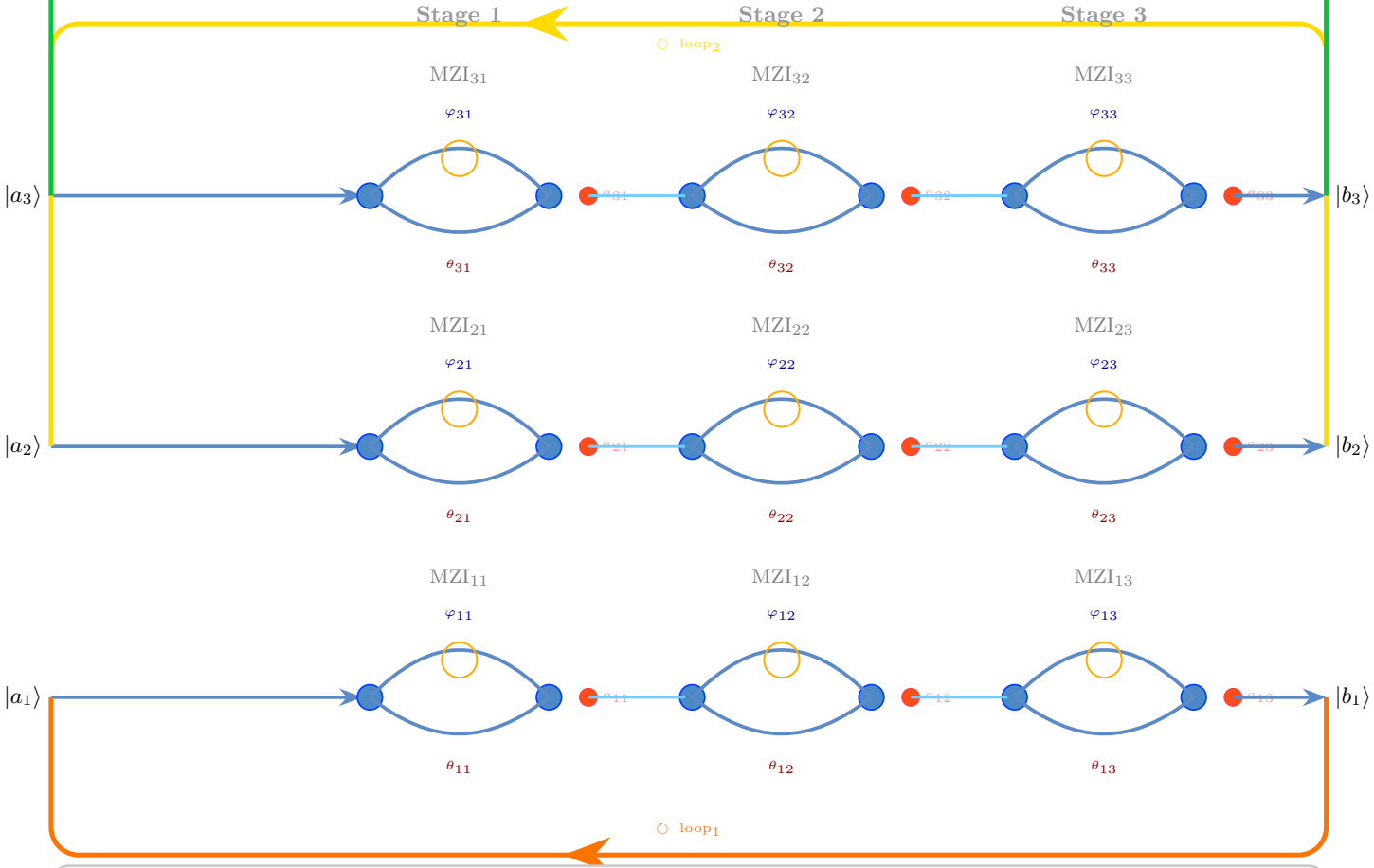


Frobenius Mesh 3×3 | Looping Photons | Complex MZI Matrix



$$A = [a_{ij}]_{3 \times 3}, \quad a_{ij} = \sin(\theta_{ij}) e^{i\varphi_{ij}}, \quad \|A\|_F = \sqrt{\sum_{i,j} |a_{ij}|^2}$$

$$A = U \Sigma V^\dagger \text{ (SVD)} \quad | \quad A^\dagger A \approx I \text{ (near-unitary)} \quad | \quad \mu \circ \delta = \text{id (IMSCRIB gate)}$$